

No.	Solution	Remarks
4(a)(i)	$\text{p.d. across LDR} = \frac{400}{400 + 8400} [4.0 - (-2.0)] = 0.27273 \text{ V}$ $\text{p.d. across LDR} = V_p - (-2.0)$ $0.27273 = V_p + 2.0$ $V_p = 0.27273 - 2.0$ $\approx -1.7 \text{ V}$	[1] for p.d. across LDR [1] for correct answer
4(a)(ii)	When a burglar <u>blocks the laser beam</u>, the <u>resistance of the LDR will be high (or increases)</u>. Hence the <u>p.d across the LDR and alarm will be high (or increases)</u>. This will turn on the alarm.	[1] [1]
4(b)(i)	There will be no deflection in the galvanometer when the p.d across JY = 1.5 V. Let the balance length (between J and Y) be L_{JY}. Then $V_{JY} = \frac{L_{JY}}{1.00} \left(\frac{1.6}{1.6 + 0.50} \right) (4.0)$ $1.5 = \frac{L_{JY}}{1.00} \left(\frac{1.6}{1.6 + 0.50} \right) (4.0)$ $L_{JY} = 0.4922$ $\approx 0.49 \text{ m}$	[1] for correct substitution [1] for correct answer
4(b)(ii)	When 1.5 Ω resistor is connected across points A and B, the terminal p.d across cell E is $V_{\text{terminal}} = \frac{1.5}{1.5 + 0.50} (1.5)$ $= 1.125 \text{ V}$ There will be null deflection when $V'_{JY} = 1.125 \text{ V}$ $V'_{JY} = \frac{L'_{JY}}{1.00} \left(\frac{1.6}{1.6 + 0.50} \right) (4.0)$ $1.125 = \frac{L'_{JY}}{1.00} \left(\frac{1.6}{1.6 + 0.50} \right) (4.0)$ $L'_{JY} = 0.3691$ $\approx 0.37 \text{ m}$	[1] for correct V_{terminal} [1] for correct substitution [1] for correct answer
5(a)	Gain in kinetic energy = Loss in electric potential energy	[1] statement